

GLM-5.2 frontier pilot v1

Status: Not supported — frontier not located. Frozen before the first GLM-5.2 call; completed on 12 July 2026.

Question

Can a small, provider-pinned pilot locate a region where GLM-5.2 produces both correct and silently wrong completed answers on the existing satisfiable 3-SAT instrument? This pilot locates a cohort; it does not test a confidence model.

Fixed call conditions

- Model: z-ai/glm-5.2 through OpenRouter.
- Provider: StreamLake only; fallback disabled.
- Prompt version: 1.
- Variables: 20.
- Completion cap: 32,768 tokens.
- Temperature: 0.6.
- Reasoning effort: high, returned reasoning retained separately.
- Correctness: mechanical verification of the final 20-bit assignment.
- Outcomes: correct completion, silently wrong completion, or loud failure.

Two-stage design

1. **Scout:** two fresh instances at each difficulty 4.2, 4.8, 5.4, 6.0, and 6.6; ten calls total.
2. **Focus:** after reporting the scout outcomes, choose three levels centered on the observed transition and run ten fresh instances per level; thirty calls total. If the scout has no transition, shift the focus range in the direction indicated by the outcomes and document that choice before calls.

The scout has a \$0.75 hard cap. The complete pilot has a \$3.00 hard cap and a 40-call cap. Provider mismatch stops collection immediately. No confidence coefficients are fit until the pilot is complete.

Readout

The pilot succeeds only as cohort discovery if at least two difficulty levels contain both correct and silently wrong completed calls, with at least eight of each label overall. Otherwise the result is `Frontier not located`, and a confidence experiment remains unauthorized.

Reasoning text, completion/reasoning tokens, elapsed time, finish reason, and provider-reported cost are retained for later feature specification. This exploratory pilot cannot establish that any of them predict correctness.

Result

The scout stopped at its ten-call cap after spending USD 0.281850. All ten calls were pinned to StreamLake with no fallback or provider mismatch.

Difficulty	Correct	Silent error	Loud failure
4.2	2	0	0
4.8	1	0	1
5.4	0	0	2
6.0	1	0	1
6.6	0	0	2

The six loud failures were four token-cap truncations, one 19-bit answer where 20 bits were required, and one malformed API response. There were no silently wrong completed answers and no mixed completed-outcome level. The cohort gate failed, so the thirty-call focus stage was not run.

This is not evidence that GLM-5.2 lacks a confidence signal. It shows that under the frozen 32,768-token, high-reasoning condition, loud reliability failures dominate before this instrument reveals a silent-error frontier.